```
import cv2 as cv
import matplotlib as plt
import numpy as np

a = cv.imread('rose.jpg', 0)
img = a.copy()           # copy the image

h = a[0:3, 0:3]          # crop a patch from the image
print(h)
print(img[0, 10])        # Access to pixel in coodinate (0, 10)
cv.imshow("Image", img)
cv.waitKey(0);
```

```
import numpy as np
import cv2
from matplotlib import pyplot as plt

### gray and binary transformation
a=cv2.imread("rose.jpg",1)
gray = cv2.cvtColor(a, cv2.COLOR_BGR2GRAY)
cv2.imshow('image',gray)
cv2.waitKey()

_, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)    # binary thresholding at 127 gray level
                                                                # with only two values (either 0 or 255)

cv2.imshow('Binary Image', binary)
###
# Convert the grayscale image to double precision

double_gray = gray.astype(np.float64)
double_gray=double_gray/255
double_gray = double_gray**0.5
print(np.max(double_gray))
print(np.min(double_gray))

fig, axs = plt.subplots(2, 1)
# Show the original image
axs[0].imshow(gray, cmap='gray')
axs[1].imshow(double_gray, cmap='gray')
plt.show()
```